Software Concepts

Operating Systems vs Application Software
Device Drivers • Utilities
Files & Storage Concepts

BCA 1st Semester

Learning Objectives

- Understand the distinction between Operating Systems and Application Software
- Learn the role and function of Device Drivers
- Explore System Utilities and their applications
- Master fundamental file and storage management concepts
- Understand file permissions and access control basics

OS vs Application Software

Operating System (OS)

Definition :

A program that acts as an intermediary between a user and the computer hardware.

Key Functions:

- Manages hardware resources
- Controls program execution
- Allocates memory
- Manages I/O operations

Application Software

Definition :

Programs designed to perform specific tasks for the user.

Characteristics:

- Runs on top of OS
- Requires OS to function
- Provides user utilities
- Examples: Word, Excel, Chrome

Key Differences

Aspect	Operating System	Application Software
Purpose	Manage hardware & resources	Perform specific user tasks
Dependency	Independent, fundamental	Depends on OS
Examples	Windows, Linux, macOS, Android	Word, Excel, Photoshop, VLC
User Interaction	Direct & Indirect	Direct (End User)

Device Drivers

Definition :

A device driver is a specialized software program that allows the OS to communicate with hardware devices. It provides the interface between application software and physical hardware components.

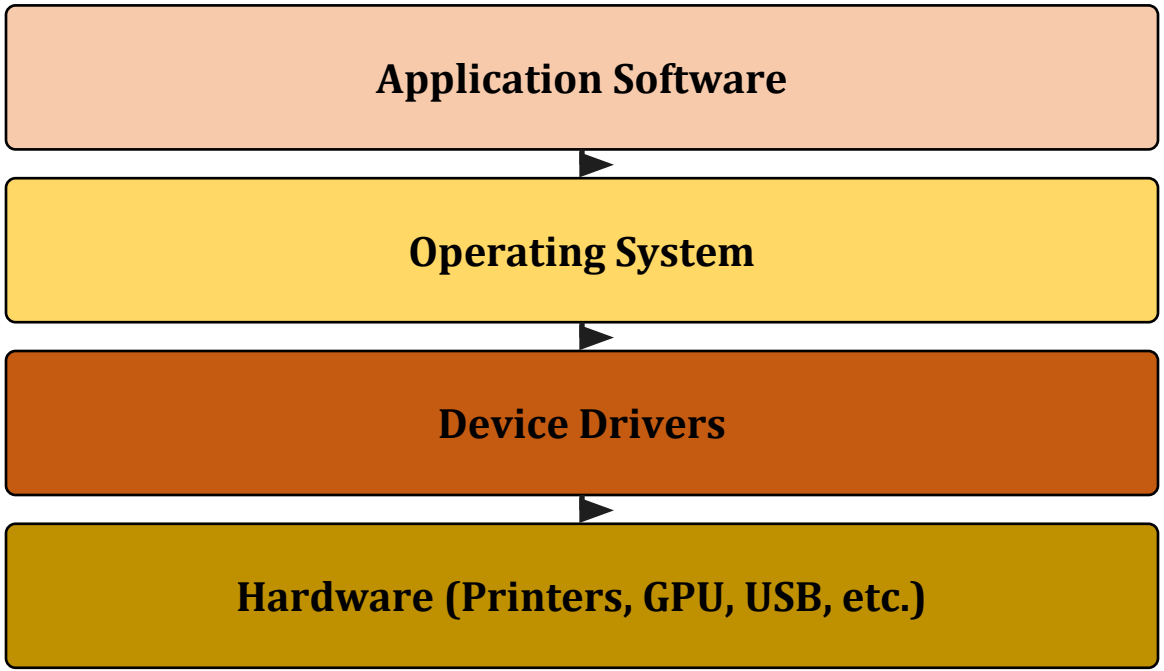
Purpose & Functions:

- Translate commands from OS to device language
- Control hardware operations
- Manage data transfer
- Enable plug-and-play functionality

Common Examples:

- Printer Driver
- Graphics/GPU Driver
- Network Driver
- USB Device Driver
- Motherboard/Chipset Driver

Device Driver Architecture



The driver translates OS commands into hardware-specific instructions and manages data exchange between the operating system and physical devices.

System Utilities

Definition : Utilities are specialized programs that enhance the functionality of the operating system and help in performing routine tasks and system maintenance.

File Management

File explorer
Create Files/Folders Copy,
Move and Delete Files
Rename Files
Search Files

Compression

WinRAR
7-Zip
WinZip

Backup & Recovery

Backup software
Data recovery
Archive tools
Data Recovery Software

Disk Utilities

Disk Defragmenter
Disk Cleanup
Partition Manager

Security Utilities

Antivirus
Firewall
Encryption tools

System Monitoring

Task Manager
System Monitor
Performance tools

Category	Examples	Purpose
File Management	File Explorer	Organize, copy, move, rename, delete files
Compression	WinRAR, WinZip, 7-Zip	Reduce file size
Backup & Recovery	Windows Backup, OneDrive, Restore Point	Prevent and recover data loss
Disk Utilities	Disk Cleanup, Defragmenter (HDD), Partition Manager	Improve storage performance
Security Utilities	Antivirus, Firewall, Encryption	Protect the computer
System Monitoring	Task Manager, Resource Monitor, Performance Monitor	Monitor CPU, RAM, Disk usage

Files & Folders (Directories)

File

Definition:
A named collection of data stored on a storage device.
Contains:
- Actual data
- File metadata (name, type, size, date)

Folder/Directory

Definition:
A container that holds files and other folders.
Purpose:
- Organizes files
- Creates hierarchy
- Simplifies navigation

Example: Hierarchical Structure

```
My Computer
  ├─ C: Drive (Local Disk)
      ├─ Users
      │   ├─ Documents
      │   │   ├─ assignment.docx
      │   │   └─ notes.txt
      │   └─ Photos
      │       └─ vacation.jpg
      └─ Program Files
          └─ Adobe
```

File Types & Extensions

A file extension is a suffix after the filename that indicates the file type and determines which application opens it.

Category	Extensions	Examples
Document	.doc, .docx, .pdf, .txt	Word files, PDFs, Notes
Spreadsheet	.xls, .xlsx, .csv	Excel files, Data tables
Image	.jpg, .png, .gif, .bmp	Photographs, Graphics
Audio/Video	.mp3, .mp4, .avi, .mov	Music, Movies, Clips
Executable	.exe, .app, .apk	Programs, Applications
Compressed	.zip, .rar, .7z	Archives, Backups

File Paths & Naming

Absolute Path

Complete path from root drive
Example:

```
C:\Users\John\Documents\assignment
.docx
```

Starts from:
- Windows/Linux root
- Drive letter

Relative Path

Path from current location
Example:

```
./Documents/assignment.docx
```

Uses:

- ./ (current folder)
- ../ (parent folder)

Path Separators:

Windows:
`Backslash (\)  →  C:\Users\Documents`

Linux/macOS:
`Forward Slash (/)  →  /home/user/documents`

File Permissions (Introduction)

File permissions control who can access, read, modify, or execute files and folders. They are security settings that protect data from unauthorized access or modification.

Three Basic Permissions:



Read (R)

View & open file
Read file contents



Write (W)

Modify file
Delete file



Execute (X)

Run/execute file
Typically for programs

Permission Scope (Linux/Unix):

Owner
File creator or owner | **Group**
Users in the same group | **Others**
All other users

Key Takeaways

- OS manages hardware; Applications provide user functionality
- Device Drivers enable OS-hardware communication
- System Utilities enhance OS functionality and system maintenance
- Files and Folders organize data in hierarchical structures
- File paths can be absolute or relative
- Permissions control access and protect data security

Thank You!

Questions & Discussion